

Results of FMEDA on microcontrollers of the STM32F4 series

Introduction

This document reports the numeric results of the safety analysis carried out on the microcontrollers of the [STM32F4 series](#). The reported metrics are:

- Relative safety metrics:
 - Diagnostic coverage (DC)
 - Safe failure fraction (SFF)
- Absolute safety metrics:
 - Probability of failure per hour (PFH)

As detailed in [1] available on www.st.com, [STM32F4 series](#) covers many part numbers. Therefore, FMEDA (failure mode effect diagnostic analysis) results are delivered to end users as static snapshots, computed for a given set of assumptions.

This document, which is related to the [X-CUBE-STL](#) software, is updated whenever new specific snapshots (related to different part numbers) are computed and available. The addition of new snapshots is reflected in the document's revision history.





1 General information

This document reports the numeric results of the safety analysis (FMEDA) carried out on the Arm® Cortex®-M4-based microcontrollers of the [STM32F4 series](#).

Caution: It must not be copied in an abridged version in the end-user documentation.

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2 Terms and abbreviations

Table 1. Terms and abbreviations

Acronym	Definition
Application software	Part of the software executed by the MCU that ensures the functionality of the end-user application and integrates the safety functions
CCF	Common cause failure
CM	Continuous mode
COTS	Commercial off-the-shelf
CoU	Conditions of use
CPU	Central processing unit
DC	Diagnostic coverage
DTI	Diagnostic test interval
End user	Individual person or company who integrates the MCU in an application, such as an electronic control board
EUC	Equipment under control
FIT	Failure in time
FMEA	Failure mode effect analysis
FMEDA	Failure mode effect diagnostic analysis
HD	High demand
HFT	Hardware fault tolerance
HW	Hardware
ITRS	International technology roadmap for semiconductors
LD	Low demand
MCU	Microcontroller unit
PFD	Probability of dangerous failure on demand
PFH	Probability of failure per hour
PST	Process safety time
SFF	Safe failure fraction
SIL	Safety integrity level

Notations and acronyms used, if not reported in the above table, are compliant with the functional documentation released for the [STM32F4 series](#), available on www.st.com.



3 References

This section mentions the normative and documents used as reference for writing this document.

3.1 Normative reference

This document is written in compliance with the IEC 61508 international norm for functional safety of electrical/electronic/programmable electronic safety-related systems. The version used as reference is IEC 61508-1/7:2010.

3.2 Reference documents

- [1] UM1840: STM32F4 series safety manual
- [2] AN5141: Results of FMEA on STM32F4 series microcontrollers
- [3] AN5321: How to customize the results of FMEDA on STM32 MCUs and MPUs
- [4] AN5936: X-CUBE-STL functional safety package: advanced topics



4 FMEDA process and results

This document includes numeric data of safety analysis (FMEDA) carried out on the microcontrollers of the [STM32F4 series](#).

Due to the large number of [STM32F4 series](#) part numbers, FMEDA output data are generated on demand, and individually included in this document. Data are generated for a specified part number, and under a specified set of assumptions (for example mission profile or power consumption). The shared assumptions and conditions common to safety result computations are described in [Section 4.2 List of common assumptions](#).

Details of the FMEDA snapshot structures are provided in [Section 4.3 About FMEDA snapshots](#). This includes instructions for their use in safety analysis of end-user systems.

Information on the methodology used to perform the FMEDA analysis is provided in [Section 4.1 FMEDA execution flow](#).

Note: If the part number selected for the end-user application is not yet included in this document, or if the end user needs the generation of FMEDA data based on an admissible set of assumptions (for example different mission profile), then refer to the guidance provided in [Section 4.3.2 Customization of FMEDA snapshot results](#). For further support, refer to the local ST sales contact.

4.1 FMEDA execution flow

FMEDA is carried out according to STMicroelectronics methodology guidelines for the application of the IEC 61508 safety standard to general-purpose microcontrollers and microprocessors. Detailed information on this methodology is available in the “List of evidences” section of [\[1\]](#), and in dedicated sections of [\[4\]](#). The information available in these documents is a high-level description of the FMEDA flow that helps the users to better understand the technique used.



4.2 List of common assumptions

Any FMEDA performed on a specific microcontroller of the **STM32F4 series**, to generate the snapshot included in this document, shares a common set of assumptions and conditions.

The assumptions (listed in the table below) derive from the application of STMicroelectronics methodology flow for the safety analysis of general-purpose microcontrollers, and therefore the delivery of FMEDA data generated under a different set is usually not possible.

Table 2. List of assumptions

ID	Assumptions
1.1	The base failure rates for permanent faults are computed on the basis of the mathematical model for integrated circuits described in section 7.3.1 of the IEC 62380 safety standard ⁽¹⁾ .
1.2	The base failure rate for transient failures is calculated on the basis of ITRS tables for sea level and assumed to be equal to the reference value of 1250 FIT/Mbit ⁽¹⁾ .
1.3	The FMEDA snapshots are generated assuming that the MCU is used in 1oo1 architecture (single channel) as described in [1] ⁽²⁾ .
2	Internal debug-related structures are classified as "no effect", according to IEC 61508-4, 3.6.4 ⁽³⁾ .
3	All MCU internal peripherals are considered safety-related ⁽³⁾ .
4	The MCU internal flash memory is considered safety-related and completely used. All MCU internal SRAMs are considered safety-related and completely used ⁽³⁾ .
5	Power consumption value used is equal to the maximum rating indicated in the device datasheet ⁽³⁾ .
6	Dissipation factor value used is the device datasheet rating for natural cooling ⁽³⁾ .
7	Package pins are all considered being used for safety-related applications ⁽³⁾ .
8	DC value used in FMEDA for package failure rate mitigation is fixed to the reference value of 99% ⁽⁴⁾ .
9	It is assumed that the safety mechanism rated as ++ (highly recommended) in the "List of safety recommendations" guidance table contained in section "Hardware and software diagnostics" of [1] are implemented and active. It is also assumed those mechanisms are implemented in compliance with descriptions reported in the same document.
10	The classification of the MCU failure modes as "safe" or "no effect" because of application-level considerations is not used in safety metrics (SFF) computations.
11	It is assumed that the recommendations of the MCU safety manual [1] have been duly followed in the design and implementation of the system hosting the MCU.
12	The absolute failure rates of the communication modules (such as UART) included in FMEDA snapshots do not include the failure rate contribution from the communication channel nor from the remote counterpart.

1. The base-failure rate values must be considered as reference values. They have been computed using general methods, and provided in this document for comparison with values given for other devices. Refer to [4] for further explanations and rationale.
2. Safety metrics computations for 1oo2 architectures (dual channel) must be executed by the end user on the basis of IEC 61508-6 guidance.
3. These assumptions bring to the computation of safety metrics in a conservative way, therefore fully in the spirit of the IEC 61508 recommendations.
4. Due to the complex overlap of different active safety mechanism on individual MCU peripherals, the computation of diagnostic coverage associated to the package failure rate is strongly application-dependent. Therefore, the FMEDA snapshot computations use the indicated reference DC value. End users are encouraged to execute on their own pin-level FMEDA, and to replace in computations the reference DC value with their new one.



4.3 About FMEDA snapshots

4.3.1 Snapshot information composition

All the snapshots included in this document (see [Section 5 FMEDA snapshot collection](#)) share some common characteristics on the way information is presented:

- Results are computed for a specific part number
- Safety metrics (relative and absolute) are provided separately for the two fault models (permanent and transient)
- Results are provided for a single microcontroller application (safety architecture 1oo1, see [\[1\]](#))
- Absolute metrics (λ) are expressed in FITs; the λ_{SAFE} , λ_{DD} , and λ_{DU} data computed according to IEC 61508-2 are provided in scientific notation
- Safety metrics are provided for each MCU function (corresponding to the modules reported in the MCU block diagram)
- Safety metrics per technology block (such as digital logic, NVM, analog) are not provided (because such details are not relevant for the end user)
- Package safety metrics are provided separately, to allow end-user dedicated, additional computations

For each snapshot, specific related assumptions and conditions are summarized at the beginning of the dedicated subsection.

The safety results are provided in two levels of detail:

- **MCU level:** failure rates for each available function/module of the microcontroller are collapsed together. The reported value at MCU level must be considered as illustrative only, since it is related to a hypothetical use case where all MCU internal peripherals are used for safety function implementation. Refer to [Section 4.4 Interpretation of FMEDA snapshot data](#) for more information.
- **Basic functions level:** safety metrics are provided separately for each basic function/module. Such functions basically correspond to the list of available hardware modules. Absolute and relative metrics are reported in two separate tables.

According to what is stated in [\[1\]](#), [STM32F4 series](#) MCUs are considered suitable for the implementation of safety functions up to SIL2 for a single channel. This statement needs to be further clarified to allow a correct understanding of the numbers reported in this document. [STM32F4 series](#) MCUs can be classified as an “element” according to IEC 61508-4, 3.4.5. Because IEC 61508 defines targets for relative (SFF) and absolute (PFD, PFH) metrics, to be applied to safety functions and/or subsystems, such targets are not strictly mandatory for [STM32F4 series](#) MCUs. Accordingly, the fact that the MCUs are suitable for the implementation of safety functions up to SIL2 for a single channel does not mean that SIL2 targets are achieved, but rather that MCU safety performances are compatible, under specific assumptions and conditions, with the achievement of SIL2 targets by the subsystem it belongs to. As an example, if an MCU function locally shows an SFF lower than 90%; in principle this is not a blocking issue from the IEC 61508 compliance point of view.

4.3.2 Customization of FMEDA snapshot results

Before being used in the end-user computations, a customization of the data included in the FMEDA snapshot provided in this document is usually required. Different customization use cases are possible. Some of them are not mutually exclusive:

1. Usually not all the available MCU internal peripherals and/or resources are used for the implementation of safety functions. The FMEDA data related to unused peripherals/resources can be excluded from end-user final computations; guidance on the criteria allowing the exclusion is provided in [\[1\]](#).
2. If some of the FMEDA assumptions are no more valid (one example: mission profile), the FMEDA data can be recomputed on the basis of the guidance provided on [\[3\]](#).
3. If the selected MCU part number is not addressed by one of the available snapshots, refer to [\[3\]](#) to adjust the FMEDA data to the selected package and/or memory size.

4.4 Interpretation of FMEDA snapshot data

The safety concept for using the STM32 MCUs and MPUs in functional safety applications is mainly software-based, and consequently requires modifications of the end-user application software, as described in the MCU safety manual ([\[1\]](#)). Accordingly, each safety metric value (mainly DC and SFF) reported in the FMEDA snapshot of this document **cannot be considered as an intrinsic characteristic of the MCU itself**. Instead, the safety metrics are the result of the implementation of the diagnostics required by the safety documentation, and above all by the safety manual).



5 FMEDA snapshot collection

5.1 STM32F469IIT

Mission profile

Telecom - Ground fixed switching

Table 3. Safety results (MCU level) - Absolute metrics expressed in FITs - STM32F469IIT

This table reports only illustrative data. Refer to [Section 4.3 About FMEDA snapshots](#) and [Section 4.4 Interpretation of FMEDA snapshot data](#) for details.

MCU level	Perm λ_{SAFE}	Perm λ_{DD}	Perm λ_{DU}	Trans λ_{SAFE}	Trans λ_{DD}	Trans λ_{DU}
Die only	1.040E-02	6.686E+01	6.363E+00	2.904E-01	3.535E+03	3.589E+02
Package	0.000E+00	1.479E+01	1.494E-01	0.000E+00	0.000E+00	0.000E+00
SOC total	1.040E-02	8.165E+01	6.513E+00	2.904E-01	3.535E+03	3.589E+02

Table 4. Safety results (MCU level) - Relative metrics - STM32F469IIT

This table reports only illustrative data. Refer to [Section 4.3 About FMEDA snapshots](#) and [Section 4.4 Interpretation of FMEDA snapshot data](#) for details.

MCU level	Perm DC	Perm SFFM	Trans DC	Trans SFFM
Die only	91.3%	91.3%	90.8%	90.8%
Package	99.0%	99.0%	N/A	N/A
SOC total	92.6%	92.6%	90.8%	90.8%

Table 5. Safety results (by function) - Absolute metrics - STM32F469IIT

Basic function	Perm λ_{SAFE}	Perm λ_{DD}	Perm λ_{DU}	Trans λ_{SAFE}	Trans λ_{DD}	Trans λ_{DU}
CPU SUBSYSTEM ⁽¹⁾	1.034E-02	4.595E-01	4.525E-02	2.868E-01	6.030E+00	8.595E-01
FLASH	0.000E+00	5.321E+01	5.292E+00	0.000E+00	2.261E+01	1.723E+00
SRAM	0.000E+00	9.673E+00	3.103E-01	0.000E+00	3.445E+03	3.442E+02
DMA1	0.000E+00	9.067E-02	9.999E-03	0.000E+00	3.321E+00	5.098E-01
DMA2	0.000E+00	8.884E-02	9.797E-03	0.000E+00	3.316E+00	5.040E-01
FSMC	5.428E-05	5.925E-02	5.422E-03	3.560E-03	3.334E+00	2.960E-01
CRC	0.000E+00	1.488E-03	8.712E-05	0.000E+00	8.760E-02	1.500E-02
USART1	0.000E+00	5.276E-03	2.981E-04	0.000E+00	3.211E-01	3.628E-02
USART2	0.000E+00	5.380E-03	3.042E-04	0.000E+00	3.211E-01	3.630E-02
USART3	0.000E+00	5.332E-03	3.019E-04	0.000E+00	3.211E-01	3.630E-02
USART6	0.000E+00	5.078E-03	2.876E-04	0.000E+00	3.096E-01	3.568E-02
UART4	0.000E+00	5.359E-03	3.033E-04	0.000E+00	3.211E-01	3.630E-02
UART5	0.000E+00	5.111E-03	2.887E-04	0.000E+00	3.096E-01	3.567E-02
UART7	0.000E+00	5.180E-03	2.928E-04	0.000E+00	3.108E-01	3.574E-02
UART8	0.000E+00	5.182E-03	2.934E-04	0.000E+00	3.119E-01	3.581E-02
I2C1	0.000E+00	5.639E-03	3.087E-04	0.000E+00	3.271E-01	1.740E-02
I2C2	0.000E+00	5.760E-03	3.162E-04	0.000E+00	3.317E-01	1.766E-02
I2C3	0.000E+00	5.659E-03	3.099E-04	0.000E+00	3.271E-01	1.740E-02
I2C4	0.000E+00	4.503E-03	1.867E-04	0.000E+00	3.272E-01	1.150E-02
I2C5	0.000E+00	4.495E-03	1.872E-04	0.000E+00	3.284E-01	1.156E-02



Basic function	Perm λ_{SAFE}	Perm λ_{DD}	Perm λ_{DU}	Trans λ_{SAFE}	Trans λ_{DD}	Trans λ_{DU}
TIM1	0.000E+00	1.144E-02	1.286E-04	0.000E+00	7.455E-01	1.003E-02
TIM2	0.000E+00	1.268E-02	1.430E-04	0.000E+00	8.126E-01	1.072E-02
TIM3	0.000E+00	9.173E-03	1.618E-04	0.000E+00	5.639E-01	2.776E-02
TIM4	0.000E+00	8.540E-03	9.830E-05	0.000E+00	5.626E-01	8.174E-03
TIM5	0.000E+00	1.268E-02	1.427E-04	0.000E+00	8.161E-01	1.076E-02
TIM6	0.000E+00	2.462E-03	5.952E-05	0.000E+00	1.693E-01	2.347E-02
TIM7	0.000E+00	2.181E-03	2.827E-05	0.000E+00	1.693E-01	4.149E-03
TIM8	0.000E+00	1.151E-02	1.290E-04	0.000E+00	7.468E-01	1.004E-02
TIM9	0.000E+00	4.754E-03	5.480E-05	0.000E+00	3.319E-01	5.796E-03
TIM10	0.000E+00	3.208E-03	3.905E-05	0.000E+00	2.362E-01	4.829E-03
TIM11	0.000E+00	3.258E-03	3.956E-05	0.000E+00	2.351E-01	4.817E-03
TIM12	0.000E+00	4.729E-03	5.469E-05	0.000E+00	3.318E-01	5.797E-03
TIM13	0.000E+00	3.243E-03	3.941E-05	0.000E+00	2.362E-01	4.829E-03
TIM14	0.000E+00	3.438E-03	4.175E-05	0.000E+00	2.541E-01	5.013E-03
CLOCKS ⁽²⁾	0.000E+00	8.604E-01	8.421E-02	0.000E+00	5.833E-01	7.682E-01
GPIOA	0.000E+00	5.718E-03	8.256E-05	0.000E+00	3.629E-01	2.085E-02
GPIOB	0.000E+00	5.950E-03	8.541E-05	0.000E+00	3.631E-01	2.086E-02
GPIOC	0.000E+00	6.004E-03	8.485E-05	0.000E+00	3.631E-01	2.084E-02
GIOD	0.000E+00	5.838E-03	8.373E-05	0.000E+00	3.630E-01	2.085E-02
GPIOE	0.000E+00	5.879E-03	8.398E-05	0.000E+00	3.630E-01	2.085E-02
GPIOF	0.000E+00	6.231E-03	8.687E-05	0.000E+00	3.633E-01	2.083E-02
GPIOG	0.000E+00	5.963E-03	8.458E-05	0.000E+00	3.631E-01	2.084E-02
GPIOH	0.000E+00	6.093E-03	8.549E-05	0.000E+00	3.632E-01	2.083E-02
GPIOI	0.000E+00	5.381E-03	7.848E-05	0.000E+00	3.532E-01	2.074E-02
GPIOJ	0.000E+00	6.682E-03	9.245E-05	0.000E+00	3.637E-01	2.086E-02
GPIOK	0.000E+00	6.270E-03	8.755E-05	0.000E+00	3.633E-01	2.084E-02
SPI1	0.000E+00	3.180E-03	1.305E-04	0.000E+00	2.412E-01	8.242E-03
SPI2	0.000E+00	4.485E-03	1.863E-04	0.000E+00	3.330E-01	1.165E-02
SPI3	0.000E+00	4.478E-03	1.872E-04	0.000E+00	3.272E-01	1.151E-02
SPI4	0.000E+00	3.177E-03	1.302E-04	0.000E+00	2.412E-01	8.242E-03
SPI5	0.000E+00	3.194E-03	1.312E-04	0.000E+00	2.446E-01	8.419E-03
SPI6	0.000E+00	3.193E-03	1.304E-04	0.000E+00	2.435E-01	8.358E-03
IWDG	0.000E+00	2.637E-03	7.796E-04	0.000E+00	5.649E-02	7.381E-02
WWDG	0.000E+00	5.186E-04	1.640E-04	0.000E+00	2.851E-02	1.972E-02
SDMMC	0.000E+00	2.329E-02	1.869E-03	0.000E+00	1.871E+00	3.964E-01
SAI1	0.000E+00	2.255E-02	3.090E-04	0.000E+00	1.486E+00	5.387E-02
PWR	0.000E+00	6.511E-01	2.958E-01	0.000E+00	1.110E-01	5.597E-02
RESET ⁽³⁾	0.000E+00	1.018E-01	6.447E-02	0.000E+00	1.344E-03	8.515E-04
QUADSPI	0.000E+00	2.244E-02	2.395E-03	0.000E+00	8.680E-01	1.247E-01
ADC	0.000E+00	3.294E-01	1.038E-01	0.000E+00	1.660E+00	4.546E-02
DAC	0.000E+00	1.547E-01	8.110E-03	0.000E+00	2.016E-01	8.241E-03



Basic function	Perm λ_{SAFE}	Perm λ_{DD}	Perm λ_{DU}	Trans λ_{SAFE}	Trans λ_{DD}	Trans λ_{DU}
RNG	0.000E+00	1.468E-03	4.430E-04	0.000E+00	3.848E-02	1.264E-01
bxCAN1	0.000E+00	9.570E-03	8.998E-04	0.000E+00	5.415E-01	4.989E-02
bxCAN2	0.000E+00	1.179E-02	9.923E-04	0.000E+00	6.879E-01	5.192E-02
DCMI	0.000E+00	5.965E-03	2.464E-03	0.000E+00	3.219E-01	3.393E-01
DMA2D	0.000E+00	5.925E-02	8.839E-03	0.000E+00	1.553E+00	5.677E-01
DSIHOST	0.000E+00	9.061E-02	1.277E-02	0.000E+00	3.964E+00	6.719E-01
ETHERNET	0.000E+00	9.630E-02	5.292E-03	0.000E+00	6.609E+00	6.862E-01
EXTI	0.000E+00	4.157E-03	4.618E-04	0.000E+00	3.597E-01	6.006E-02
LTDC	0.000E+00	1.220E-02	3.083E-02	0.000E+00	7.406E-02	2.745E+00
RTC	0.000E+00	6.264E-02	1.610E-02	0.000E+00	1.175E+00	5.111E-01
SYSCFG	0.000E+00	5.064E-03	7.371E-04	0.000E+00	1.418E-01	1.256E-01
USB OTG_FS	0.000E+00	2.640E-01	2.787E-02	0.000E+00	5.502E+00	1.112E+00
USB OTG_HS	0.000E+00	1.453E-01	1.429E-02	0.000E+00	7.787E+00	1.464E+00

1. Includes contributions from failures of Arm® Cortex®-M4 CPU, NVIC, MPU, Bus matrix, AHB/APB bridges, APB1, APB2 and DEBUG.
2. Includes all MCU clocks and related RCC control logic.
3. Includes related RCC control logic.

Table 6. Safety results (by function) - Relative metrics - STM32F469IIT

Basic function	Perm DC	Perm SFFM	Trans DC	Trans SFFM
CPU SUBSYSTEM ⁽¹⁾	91.0%	91.2%	87.5%	88.0%
FLASH	91.0%	91.0%	92.9%	92.9%
SRAM	96.9%	96.9%	90.9%	90.9%
DMA1	90.1%	90.1%	86.7%	86.7%
DMA2	90.1%	90.1%	86.8%	86.8%
FSMC	91.6%	91.6%	91.8%	91.9%
CRC	94.5%	94.5%	85.4%	85.4%
UART4	94.6%	94.6%	89.8%	89.8%
UART5	94.7%	94.7%	89.7%	89.7%
UART7	94.7%	94.7%	89.7%	89.7%
UART8	94.6%	94.6%	89.7%	89.7%
USART1	94.7%	94.7%	89.8%	89.8%
USART2	94.6%	94.6%	89.8%	89.8%
USART3	94.6%	94.6%	89.8%	89.8%
USART6	94.6%	94.6%	89.7%	89.7%
I2C1	94.8%	94.8%	94.9%	94.9%
I2C2	94.8%	94.8%	94.9%	94.9%
I2C3	94.8%	94.8%	94.9%	94.9%
I2C4	96.0%	96.0%	96.6%	96.6%
I2C5	96.0%	96.0%	96.6%	96.6%
TIM1	98.9%	98.9%	98.7%	98.7%
TIM2	98.9%	98.9%	98.7%	98.7%



Basic function	Perm DC	Perm SFFM	Trans DC	Trans SFFM
TIM3	98.3%	98.3%	95.3%	95.3%
TIM4	98.9%	98.9%	98.6%	98.6%
TIM5	98.9%	98.9%	98.7%	98.7%
TIM6	97.6%	97.6%	87.8%	87.8%
TIM7	98.7%	98.7%	97.6%	97.6%
TIM8	98.9%	98.9%	98.7%	98.7%
TIM9	98.9%	98.9%	98.3%	98.3%
TIM10	98.8%	98.8%	98.0%	98.0%
TIM11	98.8%	98.8%	98.0%	98.0%
TIM12	98.9%	98.9%	98.3%	98.3%
TIM13	98.8%	98.8%	98.0%	98.0%
TIM14	98.8%	98.8%	98.1%	98.1%
CLOCKS ⁽²⁾	91.1%	91.1%	43.2%	43.2%
GPIOA	98.6%	98.6%	94.6%	94.6%
GPIOB	98.6%	98.6%	94.6%	94.6%
GPIOC	98.6%	98.6%	94.6%	94.6%
GPIOD	98.6%	98.6%	94.6%	94.6%
GPIOE	98.6%	98.6%	94.6%	94.6%
GPIOF	98.6%	98.6%	94.6%	94.6%
GPIOG	98.6%	98.6%	94.6%	94.6%
GPIOH	98.6%	98.6%	94.6%	94.6%
GPIOI	98.6%	98.6%	94.5%	94.5%
GPIOJ	98.6%	98.6%	94.6%	94.6%
GPIOK	98.6%	98.6%	94.6%	94.6%
SPI1	96.1%	96.1%	96.7%	96.7%
SPI2	96.0%	96.0%	96.6%	96.6%
SPI3	96.0%	96.0%	96.6%	96.6%
SPI4	96.1%	96.1%	96.7%	96.7%
SPI5	96.1%	96.1%	96.7%	96.7%
SPI6	96.1%	96.1%	96.7%	96.7%
IWDG	77.2%	77.2%	43.4%	43.4%
WWDG	76.0%	76.0%	59.1%	59.1%
SDMMC	92.6%	92.6%	82.5%	82.5%
SAI1	98.6%	98.6%	96.5%	96.5%
PWR	68.8%	68.8%	66.5%	66.5%
RESET ⁽³⁾	61.2%	61.2%	61.2%	61.2%
QUADSPI	90.4%	90.4%	87.4%	87.4%
ADC	76.0%	76.0%	97.3%	97.3%
DAC	95.0%	95.0%	96.1%	96.1%
RNG	76.8%	76.8%	23.3%	23.3%
bxCAN1	91.4%	91.4%	91.6%	91.6%



Basic function	Perm DC	Perm SFFM	Trans DC	Trans SFFM
bxCAN2	92.2%	92.2%	93.0%	93.0%
DCMI	70.8%	70.8%	48.7%	48.7%
DMA2D	87.0%	87.0%	73.2%	73.2%
DSIHOST	87.6%	87.6%	85.5%	85.5%
ETHERNET	94.8%	94.8%	90.6%	90.6%
EXTI	90.0%	90.0%	85.7%	85.7%
LTDC	28.3%	28.3%	2.6%	2.6%
RTC	79.6%	79.6%	69.7%	69.7%
SYSCFG	87.3%	87.3%	53.0%	53.0%
USB OTG_FS	76.0%	76.0%	59.1%	59.1%
USB OTG_HS	76.0%	76.0%	59.1%	59.1%

1. Includes contributions from failures of Arm® Cortex®-M4 CPU, NVIC, MPU, Bus matrix, AHB/APB bridges, APB1, APB2 and DEBUG.
2. Includes all MCU clocks and related RCC control logic.
3. Includes related RCC control logic.

Table 7. Mission profile factors - STM32F469IIT

Die	1.14
Package	364.29

Note: Mission profile factor numbers are provided to allow the end customer recomputing absolute metrics with different, custom mission profiles (refer to [3] for details).

5.2 STM32F479BIT

Mission profile

Telecom - Ground fixed switching

Table 8. Safety results (MCU level) - Absolute metrics expressed in FITs - STM32F479BIT

This table reports only illustrative data. Refer to Section 4.3 About FMEDA snapshots and Section 4.4 Interpretation of FMEDA snapshot data for details.

MCU level	Perm λ_{SAFE}	Perm λ_{DD}	Perm λ_{DU}	Trans λ_{SAFE}	Trans λ_{DD}	Trans λ_{DU}
Die only	1.040E-02	6.700E+01	6.374E+00	2.904E-01	3.540E+03	3.598E+02
Package	0.000E+00	1.917E+01	1.936E-01	0.000E+00	0.000E+00	0.000E+00
SOC total	1.040E-02	8.617E+01	6.568E+00	2.904E-01	3.540E+03	3.598E+02

Table 9. Safety results (MCU level) - Relative metrics - STM32F479BIT

This table reports only illustrative data. Refer to Section 4.3 About FMEDA snapshots and Section 4.4 Interpretation of FMEDA snapshot data for details.

MCU level	Perm DC	Perm SFFM	Trans DC	Trans SFFM
Die only	91.3%	91.3%	90.8%	90.8%
Package	99.0%	99.0%	N/A	N/A
SOC total	92.9%	92.9%	90.8%	90.8%



Table 10. Safety results (by function) - Absolute metrics - STM32F479BIT

Basic function	Perm λ_{SAFE}	Perm λ_{DD}	Perm λ_{DU}	Trans λ_{SAFE}	Trans λ_{DD}	Trans λ_{DU}
CPU SUBSYSTEM ⁽¹⁾	1.034E-02	4.595E-01	4.525E-02	2.868E-01	6.030E+00	8.595E-01
FLASH	0.000E+00	5.321E+01	5.292E+00	0.000E+00	2.261E+01	1.723E+00
SRAM	0.000E+00	9.673E+00	3.103E-01	0.000E+00	3.445E+03	3.442E+02
DMA1	0.000E+00	9.067E-02	9.999E-03	0.000E+00	3.321E+00	5.098E-01
DMA2	0.000E+00	8.884E-02	9.797E-03	0.000E+00	3.316E+00	5.040E-01
FSMC	5.428E-05	5.925E-02	5.422E-03	3.560E-03	3.334E+00	2.960E-01
CRC	0.000E+00	1.488E-03	8.712E-05	0.000E+00	8.760E-02	1.500E-02
CRYP	0.000E+00	9.324E-02	5.762E-03	0.000E+00	3.020E+00	7.016E-01
HASH	0.000E+00	4.979E-02	5.532E-03	0.000E+00	1.949E+00	2.165E-01
USART1	0.000E+00	5.276E-03	2.981E-04	0.000E+00	3.211E-01	3.628E-02
USART2	0.000E+00	5.380E-03	3.042E-04	0.000E+00	3.211E-01	3.630E-02
USART3	0.000E+00	5.332E-03	3.019E-04	0.000E+00	3.211E-01	3.630E-02
USART6	0.000E+00	5.078E-03	2.876E-04	0.000E+00	3.096E-01	3.568E-02
UART4	0.000E+00	5.359E-03	3.033E-04	0.000E+00	3.211E-01	3.630E-02
UART5	0.000E+00	5.111E-03	2.887E-04	0.000E+00	3.096E-01	3.567E-02
UART7	0.000E+00	5.180E-03	2.928E-04	0.000E+00	3.108E-01	3.574E-02
UART8	0.000E+00	5.182E-03	2.934E-04	0.000E+00	3.119E-01	3.581E-02
I2C1	0.000E+00	5.639E-03	3.087E-04	0.000E+00	3.271E-01	1.740E-02
I2C2	0.000E+00	5.760E-03	3.162E-04	0.000E+00	3.317E-01	1.766E-02
I2C3	0.000E+00	5.659E-03	3.099E-04	0.000E+00	3.271E-01	1.740E-02
I2C4	0.000E+00	4.503E-03	1.867E-04	0.000E+00	3.272E-01	1.150E-02
I2C5	0.000E+00	4.495E-03	1.872E-04	0.000E+00	3.284E-01	1.156E-02
TIM1	0.000E+00	1.144E-02	1.286E-04	0.000E+00	7.455E-01	1.003E-02
TIM2	0.000E+00	1.268E-02	1.430E-04	0.000E+00	8.126E-01	1.072E-02
TIM3	0.000E+00	9.173E-03	1.618E-04	0.000E+00	5.639E-01	2.776E-02
TIM4	0.000E+00	8.540E-03	9.830E-05	0.000E+00	5.626E-01	8.174E-03
TIM5	0.000E+00	1.268E-02	1.427E-04	0.000E+00	8.161E-01	1.076E-02
TIM6	0.000E+00	2.462E-03	5.952E-05	0.000E+00	1.693E-01	2.347E-02
TIM7	0.000E+00	2.181E-03	2.827E-05	0.000E+00	1.693E-01	4.149E-03
TIM8	0.000E+00	1.151E-02	1.290E-04	0.000E+00	7.468E-01	1.004E-02
TIM9	0.000E+00	4.754E-03	5.480E-05	0.000E+00	3.319E-01	5.796E-03
TIM10	0.000E+00	3.208E-03	3.905E-05	0.000E+00	2.362E-01	4.829E-03
TIM11	0.000E+00	3.258E-03	3.956E-05	0.000E+00	2.351E-01	4.817E-03
TIM12	0.000E+00	4.729E-03	5.469E-05	0.000E+00	3.318E-01	5.797E-03
TIM13	0.000E+00	3.243E-03	3.941E-05	0.000E+00	2.362E-01	4.829E-03
TIM14	0.000E+00	3.438E-03	4.175E-05	0.000E+00	2.541E-01	5.013E-03
CLOCKS ⁽²⁾	0.000E+00	8.604E-01	8.421E-02	0.000E+00	5.833E-01	7.682E-01
GPIOA	0.000E+00	5.718E-03	8.256E-05	0.000E+00	3.629E-01	2.085E-02
GPIOB	0.000E+00	5.950E-03	8.541E-05	0.000E+00	3.631E-01	2.086E-02
GPIOC	0.000E+00	6.004E-03	8.485E-05	0.000E+00	3.631E-01	2.084E-02



Basic function	Perm λ_{SAFE}	Perm λ_{DD}	Perm λ_{DU}	Trans λ_{SAFE}	Trans λ_{DD}	Trans λ_{DU}
GPIOA	0.000E+00	5.838E-03	8.373E-05	0.000E+00	3.630E-01	2.085E-02
GPIOB	0.000E+00	5.879E-03	8.398E-05	0.000E+00	3.630E-01	2.085E-02
GPIOC	0.000E+00	6.231E-03	8.687E-05	0.000E+00	3.633E-01	2.083E-02
GPIOD	0.000E+00	5.963E-03	8.458E-05	0.000E+00	3.631E-01	2.084E-02
GPIOE	0.000E+00	6.093E-03	8.549E-05	0.000E+00	3.632E-01	2.083E-02
GPIOF	0.000E+00	5.381E-03	7.848E-05	0.000E+00	3.532E-01	2.074E-02
GPIOG	0.000E+00	6.682E-03	9.245E-05	0.000E+00	3.637E-01	2.086E-02
GPIOH	0.000E+00	6.270E-03	8.755E-05	0.000E+00	3.633E-01	2.084E-02
GPIOI	0.000E+00	3.180E-03	1.305E-04	0.000E+00	2.412E-01	8.242E-03
GPIOJ	0.000E+00	4.485E-03	1.863E-04	0.000E+00	3.330E-01	1.165E-02
GPION	0.000E+00	4.478E-03	1.872E-04	0.000E+00	3.272E-01	1.151E-02
GPIOO	0.000E+00	3.177E-03	1.302E-04	0.000E+00	2.412E-01	8.242E-03
GPIOP	0.000E+00	3.194E-03	1.312E-04	0.000E+00	2.446E-01	8.419E-03
GPIOQ	0.000E+00	3.193E-03	1.304E-04	0.000E+00	2.435E-01	8.358E-03
IWDG	0.000E+00	2.637E-03	7.796E-04	0.000E+00	5.649E-02	7.381E-02
WWDG	0.000E+00	5.186E-04	1.640E-04	0.000E+00	2.851E-02	1.972E-02
SDMMC	0.000E+00	2.329E-02	1.869E-03	0.000E+00	1.871E+00	3.964E-01
SAI1	0.000E+00	2.255E-02	3.090E-04	0.000E+00	1.486E+00	5.387E-02
PWR	0.000E+00	6.511E-01	2.958E-01	0.000E+00	1.110E-01	5.597E-02
RESET ⁽³⁾	0.000E+00	1.018E-01	6.447E-02	0.000E+00	1.344E-03	8.515E-04
QUADSPI	0.000E+00	2.244E-02	2.395E-03	0.000E+00	8.680E-01	1.247E-01
ADC	0.000E+00	3.294E-01	1.038E-01	0.000E+00	1.660E+00	4.546E-02
DAC	0.000E+00	1.547E-01	8.110E-03	0.000E+00	2.016E-01	8.241E-03
RNG	0.000E+00	1.468E-03	4.430E-04	0.000E+00	3.848E-02	1.264E-01
bxCAN1	0.000E+00	9.570E-03	8.998E-04	0.000E+00	5.415E-01	4.989E-02
bxCAN2	0.000E+00	1.179E-02	9.923E-04	0.000E+00	6.879E-01	5.192E-02
DCMI	0.000E+00	5.965E-03	2.464E-03	0.000E+00	3.219E-01	3.393E-01
DMA2D	0.000E+00	5.925E-02	8.839E-03	0.000E+00	1.553E+00	5.677E-01
DSIHOST	0.000E+00	9.061E-02	1.277E-02	0.000E+00	3.964E+00	6.719E-01
ETHERNET	0.000E+00	9.630E-02	5.292E-03	0.000E+00	6.609E+00	6.862E-01
EXTI	0.000E+00	4.157E-03	4.618E-04	0.000E+00	3.597E-01	6.006E-02
LTDC	0.000E+00	1.220E-02	3.083E-02	0.000E+00	7.406E-02	2.745E+00
RTC	0.000E+00	6.264E-02	1.610E-02	0.000E+00	1.175E+00	5.111E-01
SYSCFG	0.000E+00	5.064E-03	7.371E-04	0.000E+00	1.418E-01	1.256E-01
USB OTG_FS	0.000E+00	2.640E-01	2.787E-02	0.000E+00	5.502E+00	1.112E+00
USB OTG_HS	0.000E+00	1.453E-01	1.429E-02	0.000E+00	7.787E+00	1.464E+00

1. Includes contributions from failures of Arm® Cortex® -M4 CPU, NVIC, MPU, Bus matrix, AHB/APB bridges, APB1, APB2 and DEBUG.
2. Includes all MCU clocks and related RCC control logic.
3. Includes related RCC control logic.


Table 11. Safety results (by function) - Relative metrics - STM32F479BIT

Basic function	Perm DC	Perm SFFM	Trans DC	Trans SFFM
CPU SUBSYSTEM ⁽¹⁾	91.0%	91.2%	87.5%	88.0%
FLASH	91.0%	91.0%	92.9%	92.9%
SRAM	96.9%	96.9%	90.9%	90.9%
DMA1	90.1%	90.1%	86.7%	86.7%
DMA2	90.1%	90.1%	86.8%	86.8%
FSMC	91.6%	91.6%	91.8%	91.9%
CRC	94.5%	94.5%	85.4%	85.4%
CRYP	94.2%	94.2%	81.1%	81.1%
HASH	90.0%	90.0%	90.0%	90.0%
USART1	94.7%	94.7%	89.8%	89.8%
USART2	94.6%	94.6%	89.8%	89.8%
USART3	94.6%	94.6%	89.8%	89.8%
USART6	94.6%	94.6%	89.7%	89.7%
UART4	94.6%	94.6%	89.8%	89.8%
UART5	94.7%	94.7%	89.7%	89.7%
UART7	94.7%	94.7%	89.7%	89.7%
UART8	94.6%	94.6%	89.7%	89.7%
I2C1	94.8%	94.8%	94.9%	94.9%
I2C2	94.8%	94.8%	94.9%	94.9%
I2C3	94.8%	94.8%	94.9%	94.9%
I2C4	96.0%	96.0%	96.6%	96.6%
I2C5	96.0%	96.0%	96.6%	96.6%
TIM1	98.9%	98.9%	98.7%	98.7%
TIM2	98.9%	98.9%	98.7%	98.7%
TIM3	98.3%	98.3%	95.3%	95.3%
TIM4	98.9%	98.9%	98.6%	98.6%
TIM5	98.9%	98.9%	98.7%	98.7%
TIM6	97.6%	97.6%	87.8%	87.8%
TIM7	98.7%	98.7%	97.6%	97.6%
TIM8	98.9%	98.9%	98.7%	98.7%
TIM9	98.9%	98.9%	98.3%	98.3%
TIM10	98.8%	98.8%	98.0%	98.0%
TIM11	98.8%	98.8%	98.0%	98.0%
TIM12	98.9%	98.9%	98.3%	98.3%
TIM13	98.8%	98.8%	98.0%	98.0%
TIM14	98.8%	98.8%	98.1%	98.1%
CLOCKS ⁽²⁾	91.1%	91.1%	43.2%	43.2%
GPIOA	98.6%	98.6%	94.6%	94.6%
GPIOB	98.6%	98.6%	94.6%	94.6%
GPIOC	98.6%	98.6%	94.6%	94.6%



Basic function	Perm DC	Perm SFFM	Trans DC	Trans SFFM
GPIOD	98.6%	98.6%	94.6%	94.6%
GPIOE	98.6%	98.6%	94.6%	94.6%
GPIOF	98.6%	98.6%	94.6%	94.6%
GPIOG	98.6%	98.6%	94.6%	94.6%
GPIOH	98.6%	98.6%	94.6%	94.6%
GPIOI	98.6%	98.6%	94.5%	94.5%
GPIOJ	98.6%	98.6%	94.6%	94.6%
GPIOK	98.6%	98.6%	94.6%	94.6%
SPI1	96.1%	96.1%	96.7%	96.7%
SPI2	96.0%	96.0%	96.6%	96.6%
SPI3	96.0%	96.0%	96.6%	96.6%
SPI4	96.1%	96.1%	96.7%	96.7%
SPI5	96.1%	96.1%	96.7%	96.7%
SPI6	96.1%	96.1%	96.7%	96.7%
IWDG	77.2%	77.2%	43.4%	43.4%
WWDG	76.0%	76.0%	59.1%	59.1%
SDMMC	92.6%	92.6%	82.5%	82.5%
SAI1	98.6%	98.6%	96.5%	96.5%
PWR	68.8%	68.8%	66.5%	66.5%
RESET ⁽³⁾	61.2%	61.2%	61.2%	61.2%
QUADSPI	90.4%	90.4%	87.4%	87.4%
ADC	76.0%	76.0%	97.3%	97.3%
DAC	95.0%	95.0%	96.1%	96.1%
RNG	76.8%	76.8%	23.3%	23.3%
bxCAN1	91.4%	91.4%	91.6%	91.6%
bxCAN2	92.2%	92.2%	93.0%	93.0%
DCMI	70.8%	70.8%	48.7%	48.7%
DMA2D	87.0%	87.0%	73.2%	73.2%
DSIHOST	87.6%	87.6%	85.5%	85.5%
ETHERNET	94.8%	94.8%	90.6%	90.6%
EXTI	90.0%	90.0%	85.7%	85.7%
LTDC	28.3%	28.3%	2.6%	2.6%
RTC	79.6%	79.6%	69.7%	69.7%
SYSCFG	87.3%	87.3%	53.0%	53.0%
USB OTG_FS	90.5%	90.5%	83.2%	83.2%
USB OTG_HS	91.0%	91.0%	84.2%	84.2%

1. Includes contributions from failures of Arm® Cortex®-M4 CPU, NVIC, MPU, Bus matrix, AHB/APB bridges, APB1, APB2 and DEBUG.
2. Includes all MCU clocks and related RCC control logic.
3. Includes related RCC control logic.



Table 12. Mission profile factors - STM32F479BIT

Die	1.14
Package	364.29

Note: Mission profile factor numbers are provided to allow the end customer recomputing absolute metrics with different, custom mission profiles (refer to [3] for details).

5.3 STM32F417ZGT

Mission profile

Telecom - Ground fixed switching

Table 13. Safety results (MCU level) - Absolute metrics expressed in FITs - STM32F417ZGT

This table reports only illustrative data. Refer to Section 4.3 About FMEDA snapshots and Section 4.4 Interpretation of FMEDA snapshot data for details.

MCU level	Perm λ_{SAFE}	Perm λ_{DD}	Perm λ_{DU}	Trans λ_{SAFE}	Trans λ_{DD}	Trans λ_{DU}
Die only	1.040E-02	6.674E+01	6.318E+00	2.904E-01	1.834E+03	1.680E+02
Package	0.000E+00	1.089E+01	1.100E-01	0.000E+00	0.000E+00	0.000E+00
SOC total	1.040E-02	7.763E+01	6.428E+00	2.904E-01	1.834E+03	1.680E+02

Table 14. Safety results (MCU level) - Relative metrics - STM32F417ZGT

This table reports only illustrative data. Refer to Section 4.3 About FMEDA snapshots and Section 4.4 Interpretation of FMEDA snapshot data for details.

MCU level	Perm DC	Perm SFFM	Trans DC	Trans SFFM
Die only	91.4%	91.4%	91.6%	91.6%
Package	99.0%	99.0%	N/A	N/A
SOC total	92.4%	92.4%	91.6%	91.6%

Table 15. Safety results (by function) - Absolute metrics - STM32F417ZGT

Basic function	Perm λ_{SAFE}	Perm λ_{DD}	Perm λ_{DU}	Trans λ_{SAFE}	Trans λ_{DD}	Trans λ_{DU}
CPU SUBSYSTEM ⁽¹⁾	1.034E-02	4.595E-01	4.525E-02	2.868E-01	6.030E+00	8.595E-01
FLASH	0.000E+00	5.319E+01	5.294E+00	0.000E+00	1.271E+01	1.623E+00
SRAM	0.000E+00	9.752E+00	3.077E-01	0.000E+00	1.758E+03	1.567E+02
DMA1	0.000E+00	9.067E-02	9.999E-03	0.000E+00	3.321E+00	5.098E-01
DMA2	0.000E+00	8.884E-02	9.797E-03	0.000E+00	3.316E+00	5.040E-01
FSMC	5.428E-05	5.925E-02	5.422E-03	3.560E-03	3.334E+00	2.960E-01
CRC	0.000E+00	1.488E-03	8.712E-05	0.000E+00	8.760E-02	1.500E-02
CRYP	0.000E+00	9.324E-02	5.762E-03	0.000E+00	3.020E+00	7.016E-01
HASH	0.000E+00	4.979E-02	5.532E-03	0.000E+00	1.949E+00	2.165E-01
USART1	0.000E+00	5.276E-03	2.981E-04	0.000E+00	3.211E-01	3.628E-02
USART2	0.000E+00	5.380E-03	3.042E-04	0.000E+00	3.211E-01	3.630E-02
USART3	0.000E+00	5.332E-03	3.019E-04	0.000E+00	3.211E-01	3.630E-02
USART6	0.000E+00	5.078E-03	2.876E-04	0.000E+00	3.096E-01	3.568E-02
UART4	0.000E+00	5.359E-03	3.033E-04	0.000E+00	3.211E-01	3.630E-02
UART5	0.000E+00	5.111E-03	2.887E-04	0.000E+00	3.096E-01	3.567E-02



Basic function	Perm λ_{SAFE}	Perm λ_{DD}	Perm λ_{DU}	Trans λ_{SAFE}	Trans λ_{DD}	Trans λ_{DU}
I2C1	0.000E+00	5.639E-03	3.087E-04	0.000E+00	3.271E-01	1.740E-02
I2C2	0.000E+00	5.760E-03	3.162E-04	0.000E+00	3.317E-01	1.766E-02
I2C3	0.000E+00	5.659E-03	3.099E-04	0.000E+00	3.271E-01	1.740E-02
TIM1	0.000E+00	1.144E-02	1.286E-04	0.000E+00	7.455E-01	1.003E-02
TIM2	0.000E+00	1.268E-02	1.430E-04	0.000E+00	8.126E-01	1.072E-02
TIM3	0.000E+00	9.173E-03	1.618E-04	0.000E+00	5.639E-01	2.776E-02
TIM4	0.000E+00	8.540E-03	9.830E-05	0.000E+00	5.626E-01	8.174E-03
TIM5	0.000E+00	1.268E-02	1.427E-04	0.000E+00	8.161E-01	1.076E-02
TIM6	0.000E+00	2.462E-03	5.952E-05	0.000E+00	1.693E-01	2.347E-02
TIM7	0.000E+00	2.181E-03	2.827E-05	0.000E+00	1.693E-01	4.149E-03
TIM8	0.000E+00	1.151E-02	1.290E-04	0.000E+00	7.468E-01	1.004E-02
TIM9	0.000E+00	4.754E-03	5.480E-05	0.000E+00	3.319E-01	5.796E-03
TIM10	0.000E+00	3.208E-03	3.905E-05	0.000E+00	2.362E-01	4.829E-03
TIM11	0.000E+00	3.258E-03	3.956E-05	0.000E+00	2.351E-01	4.817E-03
TIM12	0.000E+00	4.729E-03	5.469E-05	0.000E+00	3.318E-01	5.797E-03
TIM13	0.000E+00	3.243E-03	3.941E-05	0.000E+00	2.362E-01	4.829E-03
TIM14	0.000E+00	3.438E-03	4.175E-05	0.000E+00	2.541E-01	5.013E-03
CLOCKS ⁽²⁾	0.000E+00	8.604E-01	8.421E-02	0.000E+00	5.833E-01	7.682E-01
GPIOA	0.000E+00	5.718E-03	8.256E-05	0.000E+00	3.629E-01	2.085E-02
GPIOB	0.000E+00	5.950E-03	8.541E-05	0.000E+00	3.631E-01	2.086E-02
GPIOC	0.000E+00	6.004E-03	8.485E-05	0.000E+00	3.631E-01	2.084E-02
GPIOD	0.000E+00	5.838E-03	8.373E-05	0.000E+00	3.630E-01	2.085E-02
GPIOE	0.000E+00	5.879E-03	8.398E-05	0.000E+00	3.630E-01	2.085E-02
GPIOF	0.000E+00	6.231E-03	8.687E-05	0.000E+00	3.633E-01	2.083E-02
GPIOG	0.000E+00	5.963E-03	8.458E-05	0.000E+00	3.631E-01	2.084E-02
GPIOH	0.000E+00	6.093E-03	8.549E-05	0.000E+00	3.632E-01	2.083E-02
GPIOI	0.000E+00	5.381E-03	7.848E-05	0.000E+00	3.532E-01	2.074E-02
SPI1	0.000E+00	3.180E-03	1.305E-04	0.000E+00	2.412E-01	8.242E-03
SPI2	0.000E+00	4.485E-03	1.863E-04	0.000E+00	3.330E-01	1.165E-02
SPI3	0.000E+00	4.478E-03	1.872E-04	0.000E+00	3.272E-01	1.151E-02
IWDG	0.000E+00	2.637E-03	7.796E-04	0.000E+00	5.649E-02	7.381E-02
WWDG	0.000E+00	5.186E-04	1.640E-04	0.000E+00	2.851E-02	1.972E-02
SDMMC	0.000E+00	2.329E-02	1.869E-03	0.000E+00	1.871E+00	3.964E-01
SAI1	0.000E+00	2.255E-02	3.090E-04	0.000E+00	1.486E+00	5.387E-02
PWR	0.000E+00	6.511E-01	2.958E-01	0.000E+00	1.110E-01	5.597E-02
RESET ⁽³⁾	0.000E+00	1.018E-01	6.447E-02	0.000E+00	1.344E-03	8.515E-04
ADC	0.000E+00	3.294E-01	1.038E-01	0.000E+00	1.660E+00	4.546E-02
DAC	0.000E+00	1.547E-01	8.110E-03	0.000E+00	2.016E-01	8.241E-03
RNG	0.000E+00	1.468E-03	4.430E-04	0.000E+00	3.848E-02	1.264E-01
bxCAN1	0.000E+00	9.570E-03	8.998E-04	0.000E+00	5.415E-01	4.989E-02
bxCAN2	0.000E+00	1.179E-02	9.923E-04	0.000E+00	6.879E-01	5.192E-02



Basic function	Perm λ_{SAFE}	Perm λ_{DD}	Perm λ_{DU}	Trans λ_{SAFE}	Trans λ_{DD}	Trans λ_{DU}
DCMI	0.000E+00	5.965E-03	2.464E-03	0.000E+00	3.219E-01	3.393E-01
ETHERNET	0.000E+00	9.630E-02	5.292E-03	0.000E+00	6.609E+00	6.862E-01
RTC	0.000E+00	6.264E-02	1.610E-02	0.000E+00	1.175E+00	5.111E-01
EXTI	0.000E+00	4.157E-03	4.618E-04	0.000E+00	3.597E-01	6.006E-02
SYSCFG	0.000E+00	5.064E-03	7.371E-04	0.000E+00	1.418E-01	1.256E-01
USB OTG_FS	0.000E+00	2.640E-01	2.787E-02	0.000E+00	5.502E+00	1.112E+00
USB OTG_HS	0.000E+00	1.453E-01	1.429E-02	0.000E+00	7.787E+00	1.464E+00

1. Includes contributions from failures of Arm® Cortex®-M4 CPU, NVIC, MPU, Bus matrix, AHB/APB bridges, APB1, APB2 and DEBUG.
2. Includes all MCU clocks and related RCC control logic.
3. Includes related RCC control logic.

Table 16. Safety results (by function) - Relative metrics - STM32F417ZGT

Basic function	Perm DC	Perm SFFM	Trans DC	Trans SFFM
CPU SUBSYSTEM ⁽¹⁾	91.0%	91.2%	87.5%	88.0%
FLASH	90.9%	90.9%	88.7%	88.7%
SRAM	96.9%	96.9%	91.8%	91.8%
DMA1	90.1%	90.1%	86.7%	86.7%
DMA2	90.1%	90.1%	86.8%	86.8%
FSMC	91.6%	91.6%	91.8%	91.9%
CRC	94.5%	94.5%	85.4%	85.4%
CRYP	94.2%	94.2%	81.1%	81.1%
HASH	90.0%	90.0%	90.0%	90.0%
USART1	94.7%	94.7%	89.8%	89.8%
USART2	94.6%	94.6%	89.8%	89.8%
USART3	94.6%	94.6%	89.8%	89.8%
USART6	94.6%	94.6%	89.7%	89.7%
UART4	94.6%	94.6%	89.8%	89.8%
UART5	94.7%	94.7%	89.7%	89.7%
I2C1	94.8%	94.8%	94.9%	94.9%
I2C2	94.8%	94.8%	94.9%	94.9%
I2C3	94.8%	94.8%	94.9%	94.9%
TIM1	98.9%	98.9%	98.7%	98.7%
TIM2	98.9%	98.9%	98.7%	98.7%
TIM3	98.3%	98.3%	95.3%	95.3%
TIM4	98.9%	98.9%	98.6%	98.6%
TIM5	98.9%	98.9%	98.7%	98.7%
TIM6	97.6%	97.6%	87.8%	87.8%
TIM7	98.7%	98.7%	97.6%	97.6%
TIM8	98.9%	98.9%	98.7%	98.7%
TIM9	98.9%	98.9%	98.3%	98.3%
TIM10	98.8%	98.8%	98.0%	98.0%



Basic function	Perm DC	Perm SFFM	Trans DC	Trans SFFM
TIM11	98.8%	98.8%	98.0%	98.0%
TIM12	98.9%	98.9%	98.3%	98.3%
TIM13	98.8%	98.8%	98.0%	98.0%
TIM14	98.8%	98.8%	98.1%	98.1%
CLOCKS ⁽²⁾	91.1%	91.1%	43.2%	43.2%
GPIOA	98.6%	98.6%	94.6%	94.6%
GPIOB	98.6%	98.6%	94.6%	94.6%
GPIOC	98.6%	98.6%	94.6%	94.6%
GIOD	98.6%	98.6%	94.6%	94.6%
GPIOE	98.6%	98.6%	94.6%	94.6%
GPIOF	98.6%	98.6%	94.6%	94.6%
GPIOG	98.6%	98.6%	94.6%	94.6%
GPIOH	98.6%	98.6%	94.6%	94.6%
GPIOI	98.6%	98.6%	94.5%	94.5%
SPI1	96.1%	96.1%	96.7%	96.7%
SPI2	96.0%	96.0%	96.6%	96.6%
SPI3	96.0%	96.0%	96.6%	96.6%
IWDG	77.2%	77.2%	43.4%	43.4%
WWDG	76.0%	76.0%	59.1%	59.1%
SDMMC	92.6%	92.6%	82.5%	82.5%
SAI1	98.6%	98.6%	96.5%	96.5%
PWR	68.8%	68.8%	66.5%	66.5%
RESET ⁽³⁾	61.2%	61.2%	61.2%	61.2%
ADC	76.0%	76.0%	97.3%	97.3%
DAC	95.0%	95.0%	96.1%	96.1%
RNG	76.8%	76.8%	23.3%	23.3%
bxCAN1	91.4%	91.4%	91.6%	91.6%
bxCAN2	92.2%	92.2%	93.0%	93.0%
DCMI	70.8%	70.8%	48.7%	48.7%
ETHERNET	94.8%	94.8%	90.6%	90.6%
RTC	79.6%	79.6%	69.7%	69.7%
EXTI	90.0%	90.0%	85.7%	85.7%
SYSCFG	87.3%	87.3%	53.0%	53.0%
USB OTG_FS	90.5%	90.5%	83.2%	83.2%
USB OTG_HS	91.0%	91.0%	84.2%	84.2%

1. Includes contributions from failures of Arm® Cortex®-M4 CPU, NVIC, MPU, Bus matrix, AHB/APB bridges, APB1, APB2 and DEBUG.

2. Includes all MCU clocks and related RCC control logic.

3. Includes related RCC control logic.



Table 17. Mission profile factors - STM32F417ZGT

Die	1.14
Package	364.29

Note: Mission profile factor numbers are provided to allow the end customer recomputing absolute metrics with different, custom mission profiles (refer to [3] for details).

5.4 STM32F405VGT

Mission profile

Telecom - Ground fixed switching

Table 18. Safety results (MCU level) - Absolute metrics expressed in FITs - STM32F405VGT

This table reports only illustrative data. Refer to Section 4.3 About FMEDA snapshots and Section 4.4 Interpretation of FMEDA snapshot data for details.

MCU level	Perm λ_{SAFE}	Perm λ_{DD}	Perm λ_{DU}	Trans λ_{SAFE}	Trans λ_{DD}	Trans λ_{DU}
Die only	1.192E-02	7.619E+01	7.220E+00	2.904E-01	1.819E+03	1.659E+02
Package	0.000E+00	5.982E+00	6.042E-02	0.000E+00	0.000E+00	0.000E+00
SOC total	1.192E-02	8.217E+01	7.281E+00	2.904E-01	1.819E+03	1.659E+02

Table 19. Safety results (MCU level) - Relative metrics - STM32F405VGT

This table reports only illustrative data. Refer to Section 4.3 About FMEDA snapshots and Section 4.4 Interpretation of FMEDA snapshot data for details.

MCU level	Perm DC	Perm SFFM	Trans DC	Trans SFFM
Die only	91.3%	91.3%	91.6%	91.6%
Package	99.0%	99.0%	N/A	N/A
SOC total	91.9%	91.9%	91.6%	91.6%

Table 20. Safety results (by function) - Absolute metrics - STM32F405VGT

Basic function	Perm λ_{SAFE}	Perm λ_{DD}	Perm λ_{DU}	Trans λ_{SAFE}	Trans λ_{DD}	Trans λ_{DU}
CPU SUBSYSTEM ⁽¹⁾	1.186E-02	5.268E-01	5.187E-02	2.868E-01	6.030E+00	8.595E-01
FLASH	0.000E+00	6.097E+01	6.069E+00	0.000E+00	1.271E+01	1.623E+00
SRAM	0.000E+00	1.118E+01	3.528E-01	0.000E+00	1.758E+03	1.567E+02
DMA1	0.000E+00	1.039E-01	1.146E-02	0.000E+00	3.321E+00	5.098E-01
DMA2	0.000E+00	1.019E-01	1.123E-02	0.000E+00	3.316E+00	5.040E-01
FSMC	6.223E-05	6.792E-02	6.216E-03	3.560E-03	3.334E+00	2.960E-01
CRC	0.000E+00	1.706E-03	9.988E-05	0.000E+00	8.760E-02	1.500E-02
USART1	0.000E+00	6.049E-03	3.418E-04	0.000E+00	3.211E-01	3.628E-02
USART2	0.000E+00	6.168E-03	3.487E-04	0.000E+00	3.211E-01	3.630E-02
USART3	0.000E+00	6.113E-03	3.461E-04	0.000E+00	3.211E-01	3.630E-02
USART6	0.000E+00	5.822E-03	3.297E-04	0.000E+00	3.096E-01	3.568E-02
UART4	0.000E+00	6.143E-03	3.477E-04	0.000E+00	3.211E-01	3.630E-02
UART5	0.000E+00	5.859E-03	3.309E-04	0.000E+00	3.096E-01	3.567E-02
I2C1	0.000E+00	6.465E-03	3.539E-04	0.000E+00	3.271E-01	1.740E-02
I2C2	0.000E+00	6.603E-03	3.624E-04	0.000E+00	3.317E-01	1.766E-02



Basic function	Perm λ_{SAFE}	Perm λ_{DD}	Perm λ_{DU}	Trans λ_{SAFE}	Trans λ_{DD}	Trans λ_{DU}
I2C3	0.000E+00	6.488E-03	3.553E-04	0.000E+00	3.271E-01	1.740E-02
TIM1	0.000E+00	1.312E-02	1.475E-04	0.000E+00	7.455E-01	1.003E-02
TIM2	0.000E+00	1.454E-02	1.639E-04	0.000E+00	8.126E-01	1.072E-02
TIM3	0.000E+00	1.052E-02	1.855E-04	0.000E+00	5.639E-01	2.776E-02
TIM4	0.000E+00	9.790E-03	1.127E-04	0.000E+00	5.626E-01	8.174E-03
TIM5	0.000E+00	1.454E-02	1.636E-04	0.000E+00	8.161E-01	1.076E-02
TIM6	0.000E+00	2.823E-03	6.824E-05	0.000E+00	1.693E-01	2.347E-02
TIM7	0.000E+00	2.500E-03	3.241E-05	0.000E+00	1.693E-01	4.149E-03
TIM8	0.000E+00	1.320E-02	1.479E-04	0.000E+00	7.468E-01	1.004E-02
TIM9	0.000E+00	5.450E-03	6.283E-05	0.000E+00	3.319E-01	5.796E-03
TIM10	0.000E+00	3.678E-03	4.477E-05	0.000E+00	2.362E-01	4.829E-03
TIM11	0.000E+00	3.735E-03	4.535E-05	0.000E+00	2.351E-01	4.817E-03
TIM12	0.000E+00	5.422E-03	6.270E-05	0.000E+00	3.318E-01	5.797E-03
TIM13	0.000E+00	3.718E-03	4.518E-05	0.000E+00	2.362E-01	4.829E-03
TIM14	0.000E+00	3.942E-03	4.786E-05	0.000E+00	2.541E-01	5.013E-03
CLOCKS ⁽²⁾	0.000E+00	9.864E-01	9.654E-02	0.000E+00	5.833E-01	7.682E-01
GPIOA	0.000E+00	6.555E-03	9.464E-05	0.000E+00	3.629E-01	2.085E-02
GPIOB	0.000E+00	6.821E-03	9.792E-05	0.000E+00	3.631E-01	2.086E-02
GPIOC	0.000E+00	6.883E-03	9.727E-05	0.000E+00	3.631E-01	2.084E-02
GPIOD	0.000E+00	6.693E-03	9.599E-05	0.000E+00	3.630E-01	2.085E-02
GPIOE	0.000E+00	6.740E-03	9.627E-05	0.000E+00	3.630E-01	2.085E-02
GPIOH	0.000E+00	6.985E-03	9.801E-05	0.000E+00	3.632E-01	2.083E-02
SPI1	0.000E+00	3.646E-03	1.497E-04	0.000E+00	2.412E-01	8.242E-03
SPI2	0.000E+00	5.141E-03	2.135E-04	0.000E+00	3.330E-01	1.165E-02
SPI3	0.000E+00	5.134E-03	2.146E-04	0.000E+00	3.272E-01	1.151E-02
IWDG	0.000E+00	3.023E-03	8.937E-04	0.000E+00	5.649E-02	7.381E-02
WWDG	0.000E+00	5.945E-04	1.880E-04	0.000E+00	2.851E-02	1.972E-02
SDMMC	0.000E+00	2.670E-02	2.143E-03	0.000E+00	1.871E+00	3.964E-01
PWR	0.000E+00	7.465E-01	3.391E-01	0.000E+00	1.110E-01	5.597E-02
RESET ⁽³⁾	0.000E+00	1.167E-01	7.391E-02	0.000E+00	1.344E-03	8.515E-04
ADC	0.000E+00	3.777E-01	1.190E-01	0.000E+00	1.660E+00	4.546E-02
DAC	0.000E+00	1.773E-01	9.297E-03	0.000E+00	2.016E-01	8.241E-03
RNG	0.000E+00	1.683E-03	5.079E-04	0.000E+00	3.848E-02	1.264E-01
bxCAN1	0.000E+00	1.097E-02	1.032E-03	0.000E+00	5.415E-01	4.989E-02
bxCAN2	0.000E+00	1.352E-02	1.138E-03	0.000E+00	6.879E-01	5.192E-02
RTC	0.000E+00	7.181E-02	1.845E-02	0.000E+00	1.175E+00	5.111E-01
EXTI	0.000E+00	4.765E-03	5.295E-04	0.000E+00	3.597E-01	6.006E-02
SYSCFG	0.000E+00	5.806E-03	8.450E-04	0.000E+00	1.418E-01	1.256E-01
USB OTG_FS	0.000E+00	3.027E-01	3.195E-02	0.000E+00	5.502E+00	1.112E+00
USB OTG_HS	0.000E+00	1.666E-01	1.638E-02	0.000E+00	7.787E+00	1.464E+00



1. Includes contributions from failures of Arm® Cortex®-M4 CPU, NVIC, MPU, Bus matrix, AHB/APB bridges, APB1, APB2 and DEBUG.
2. Includes all MCU clocks and related RCC control logic.
3. Includes related RCC control logic.

Table 21. Safety results (by function) - Relative metrics - STM32F405VGT

Basic function	Perm DC	Perm SFFM	Trans DC	Trans SFFM
CPU SUBSYSTEM ⁽¹⁾	91.0%	91.2%	87.5%	88.0%
FLASH	90.9%	90.9%	88.7%	88.7%
SRAM	96.9%	96.9%	91.8%	91.8%
DMA1	90.1%	90.1%	86.7%	86.7%
DMA2	90.1%	90.1%	86.8%	86.8%
FSMC	91.6%	91.6%	91.8%	91.9%
CRC	94.5%	94.5%	85.4%	85.4%
USART1	94.7%	94.7%	89.8%	89.8%
USART2	94.6%	94.6%	89.8%	89.8%
USART3	94.6%	94.6%	89.8%	89.8%
USART6	94.6%	94.6%	89.7%	89.7%
UART4	94.6%	94.6%	89.8%	89.8%
UART5	94.7%	94.7%	89.7%	89.7%
I2C1	94.8%	94.8%	94.9%	94.9%
I2C2	94.8%	94.8%	94.9%	94.9%
I2C3	94.8%	94.8%	94.9%	94.9%
TIM1	98.9%	98.9%	98.7%	98.7%
TIM2	98.9%	98.9%	98.7%	98.7%
TIM3	98.3%	98.3%	95.3%	95.3%
TIM4	98.9%	98.9%	98.6%	98.6%
TIM5	98.9%	98.9%	98.7%	98.7%
TIM6	97.6%	97.6%	87.8%	87.8%
TIM7	98.7%	98.7%	97.6%	97.6%
TIM8	98.9%	98.9%	98.7%	98.7%
TIM9	98.9%	98.9%	98.3%	98.3%
TIM10	98.8%	98.8%	98.0%	98.0%
TIM11	98.8%	98.8%	98.0%	98.0%
TIM12	98.9%	98.9%	98.3%	98.3%
TIM13	98.8%	98.8%	98.0%	98.0%
TIM14	98.8%	98.8%	98.1%	98.1%
CLOCKS ⁽²⁾	91.1%	91.1%	43.2%	43.2%
GPIOA	98.6%	98.6%	94.6%	94.6%
GPIOB	98.6%	98.6%	94.6%	94.6%
GPIOC	98.6%	98.6%	94.6%	94.6%
GPIOD	98.6%	98.6%	94.6%	94.6%
GPIOE	98.6%	98.6%	94.6%	94.6%



Basic function	Perm DC	Perm SFFM	Trans DC	Trans SFFM
GPIOH	98.6%	98.6%	94.6%	94.6%
SPI1	96.1%	96.1%	96.7%	96.7%
SPI2	96.0%	96.0%	96.6%	96.6%
SPI3	96.0%	96.0%	96.6%	96.6%
IWDG	77.2%	77.2%	43.4%	43.4%
WWDG	76.0%	76.0%	59.1%	59.1%
SDMMC	92.6%	92.6%	82.5%	82.5%
PWR	68.8%	68.8%	66.5%	66.5%
RESET ⁽³⁾	61.2%	61.2%	61.2%	61.2%
ADC	76.0%	76.0%	97.3%	97.3%
DAC	95.0%	95.0%	96.1%	96.1%
RNG	76.8%	76.8%	23.3%	23.3%
bxCAN1	91.4%	91.4%	91.6%	91.6%
bxCAN2	92.2%	92.2%	93.0%	93.0%
RTC	79.6%	79.6%	69.7%	69.7%
EXTI	90.0%	90.0%	85.7%	85.7%
SYSCFG	87.3%	87.3%	53.0%	53.0%
USB OTG_FS	90.5%	90.5%	83.2%	83.2%
USB OTG_HS	91.0%	91.0%	84.2%	84.2%

1. Includes contributions from failures of Arm® Cortex®-M4 CPU, NVIC, MPU, Bus matrix, AHB/APB bridges, APB1, APB2 and DEBUG.
2. Includes all MCU clocks and related RCC control logic.
3. Includes related RCC control logic.

Table 22. Mission profile factors - STM32F405VGT

Die	1.31
Package	364.29

Note: Mission profile factor numbers are provided to allow the end customer recomputing absolute metrics with different, custom mission profiles (refer to [3] for details).

5.5 STM32F410CBT

Mission profile

Telecom - Ground fixed switching

Table 23. Safety results (MCU level) - Absolute metrics expressed in FITs - STM32F410CBT

This table reports only illustrative data. Refer to Section 4.3 About FMEDA snapshots and Section 4.4 Interpretation of FMEDA snapshot data for details.

Absolute metrics	Perm λSAFE	Perm λDD	Perm λDU	Trans λSAFE	Trans λDD	Trans λDU
Die only	2.612E-02	9.029E+01	4.446E+00	2.868E-01	3.237E+02	2.119E+01
Package	0.000E+00	1.867E+00	1.886E-02	0.000E+00	0.000E+00	0.000E+00
SOC total	2.612E-02	9.216E+01	4.465E+00	2.868E-01	3.237E+02	2.119E+01

**Table 24. Safety results (MCU level) - Relative metrics - STM32F410CBT**

This table reports only illustrative data. Refer to [Section 4.3 About FMEDA snapshots](#) and [Section 4.4 Interpretation of FMEDA snapshot data](#) for details.

MCU level	Perm DC	Perm SFFM	Trans DC	Trans SFFM
Die only	95.3%	95.3%	93.9%	93.9%
Package	99.0%	99.0%	N/A	N/A
SOC total	95.4%	95.4%	93.9%	93.9%

Table 25. Safety results (by function) - Absolute metrics - STM32F410CBT

Module	Perm λ_{SAFE}	Perm λ_{DD}	Perm λ_{DU}	Trans λ_{SAFE}	Trans λ_{DD}	Trans λ_{DU}
CPU SUBSYSTEM ⁽¹⁾	2.612E-02	8.658E-01	9.862E-02	2.868E-01	6.172E+00	9.851E-01
CRC	0.000E+00	3.758E-03	2.200E-04	0.000E+00	8.760E-02	1.500E-02
ADC ⁽²⁾	0.000E+00	4.182E-01	1.343E-01	0.000E+00	5.560E-01	1.605E-02
GPIO-PORTH	0.000E+00	7.370E-03	1.354E-04	0.000E+00	1.199E-01	1.838E-02
GPIO-PORTC	0.000E+00	1.516E-02	2.142E-04	0.000E+00	3.631E-01	2.084E-02
GPIO-PORTB	0.000E+00	1.502E-02	2.157E-04	0.000E+00	3.631E-01	2.086E-02
GPIO-PORTA	0.000E+00	1.444E-02	2.084E-04	0.000E+00	3.629E-01	2.085E-02
RNG	0.000E+00	3.708E-03	1.119E-03	0.000E+00	3.848E-02	1.264E-01
SRAM	0.000E+00	1.269E+01	4.063E-01	0.000E+00	2.971E+02	1.565E+01
DAC	0.000E+00	2.054E-01	1.073E-02	0.000E+00	2.016E-01	8.241E-03
TIM6	0.000E+00	6.216E-03	1.503E-04	0.000E+00	1.693E-01	2.347E-02
LPTIM1	0.000E+00	6.216E-03	1.503E-04	0.000E+00	1.693E-01	2.347E-02
TIM5	0.000E+00	3.203E-02	3.602E-04	0.000E+00	8.161E-01	1.076E-02
TIM9	0.000E+00	8.101E-03	9.861E-05	0.000E+00	2.362E-01	4.829E-03
TIM11	0.000E+00	8.226E-03	9.988E-05	0.000E+00	2.351E-01	4.817E-03
TIM1	0.000E+00	2.889E-02	3.248E-04	0.000E+00	7.455E-01	1.003E-02
I2C1	0.000E+00	1.424E-02	7.795E-04	0.000E+00	3.271E-01	1.740E-02
I2C2	0.000E+00	1.454E-02	7.983E-04	0.000E+00	3.317E-01	1.766E-02
I2C4	0.000E+00	1.137E-02	4.715E-04	0.000E+00	3.272E-01	1.150E-02
RTC	0.000E+00	1.302E-01	3.754E-02	0.000E+00	1.175E+00	5.111E-01
USART1	0.000E+00	1.353E-02	7.658E-04	0.000E+00	3.211E-01	3.630E-02
USART2	0.000E+00	1.358E-02	7.680E-04	0.000E+00	3.211E-01	3.630E-02
USART6	0.000E+00	1.308E-02	7.393E-04	0.000E+00	3.108E-01	3.574E-02
SPI1	0.000E+00	8.030E-03	3.296E-04	0.000E+00	2.412E-01	8.242E-03
SPI2	0.000E+00	1.132E-02	4.703E-04	0.000E+00	3.330E-01	1.165E-02
SPI3	0.000E+00	1.131E-02	4.726E-04	0.000E+00	3.272E-01	1.151E-02
SPI5	0.000E+00	3.209E-03	7.569E-05	0.000E+00	1.083E-01	1.245E-03
IWDG	0.000E+00	6.659E-03	1.968E-03	0.000E+00	5.649E-02	7.381E-02
WWDG	0.000E+00	1.309E-03	4.140E-04	0.000E+00	2.851E-02	1.972E-02
FLASH	0.000E+00	7.312E+01	3.113E+00	0.000E+00	4.051E+00	1.536E+00
DMA1	0.000E+00	2.289E-01	2.525E-02	0.000E+00	3.321E+00	5.098E-01
DMA2	0.000E+00	2.243E-01	2.474E-02	0.000E+00	3.316E+00	5.040E-01
CLOCK ⁽³⁾	0.000E+00	1.149E+00	1.134E-01	0.000E+00	5.833E-01	7.682E-01



Module	Perm λ_{SAFE}	Perm λ_{DD}	Perm λ_{DU}	Trans λ_{SAFE}	Trans λ_{DD}	Trans λ_{DU}
EXTI	0.000E+00	1.050E-02	1.166E-03	0.000E+00	3.597E-01	6.006E-02
POWER	0.000E+00	8.504E-01	3.855E-01	0.000E+00	1.110E-01	5.597E-02
RESET ⁽⁴⁾	0.000E+00	1.325E-01	8.395E-02	0.000E+00	1.344E-03	8.515E-04

1. Includes contributions from failures of Arm® Cortex®-M4 CPU, NVIC, MPU, Bus matrix, AHB/APB bridges, APB1, APB2, SYSCFG, and DEBUG.
2. Includes temperature sensor.
3. Includes all MCU clocks and related RCC control logic.
4. Includes related RCC control logic.

Table 26. Safety results (by function) - Relative metrics - STM32F410CBT

Module	Perm DC	Perm SFFM	Trans DC	Trans SFFM
CPU SUBSYSTEM ⁽¹⁾	89.8%	90.0%	86.2%	86.8%
CRC	94.5%	94.5%	85.4%	85.4%
ADC ⁽²⁾	75.7%	75.7%	97.2%	97.2%
GPIO-PORTH	98.2%	98.2%	86.7%	86.7%
GPIO-PORTC	98.6%	98.6%	94.6%	94.6%
GPIO-PORTB	98.6%	98.6%	94.6%	94.6%
GPIO-PORTA	98.6%	98.6%	94.6%	94.6%
RNG	76.8%	76.8%	23.3%	23.3%
SRAM	96.9%	96.9%	95.0%	95.0%
DAC	95.0%	95.0%	96.1%	96.1%
TIM6	97.6%	97.6%	87.8%	87.8%
LPTIM1	97.6%	97.6%	87.8%	87.8%
TIM5	98.9%	98.9%	98.7%	98.7%
TIM9	98.8%	98.8%	98.0%	98.0%
TIM11	98.8%	98.8%	98.0%	98.0%
TIM1	98.9%	98.9%	98.7%	98.7%
I2C1	94.8%	94.8%	94.9%	94.9%
I2C2	94.8%	94.8%	94.9%	94.9%
I2C4	96.0%	96.0%	96.6%	96.6%
RTC	77.6%	77.6%	69.7%	69.7%
USART1	94.6%	94.6%	89.8%	89.8%
USART2	94.6%	94.6%	89.8%	89.8%
USART6	94.7%	94.7%	89.7%	89.7%
SPI1	96.1%	96.1%	96.7%	96.7%
SPI2	96.0%	96.0%	96.6%	96.6%
SPI3	96.0%	96.0%	96.6%	96.6%
SPI5	97.7%	97.7%	98.9%	98.9%
IWDG	77.2%	77.2%	43.4%	43.4%
WWDG	76.0%	76.0%	59.1%	59.1%
FLASH	95.9%	95.9%	72.5%	72.5%
DMA1	90.1%	90.1%	86.7%	86.7%



Module	Perm DC	Perm SFFM	Trans DC	Trans SFFM
DMA2	90.1%	90.1%	86.8%	86.8%
CLOCK ⁽³⁾	91.0%	91.0%	43.2%	43.2%
EXTI	90.0%	90.0%	85.7%	85.7%
POWER	68.8%	68.8%	66.5%	66.5%
RESET ⁽⁴⁾	61.2%	61.2%	61.2%	61.2%

1. Includes contributions from failures of Arm® Cortex®-M4 CPU, NVIC, MPU, Bus matrix, AHB/APB bridges, APB1, APB2, SYSCFG, and DEBUG.
2. Includes temperature sensor.
3. Includes all MCU clocks and related RCC control logic.
4. Includes related RCC control logic.

Table 27. Mission profile factors - STM32F410CBT

Die	1.49
Package	364.29

Note:

Mission profile factor numbers are provided to allow the end customer recomputing absolute metrics with different, custom mission profiles (refer to [3] for details).



Revision history

Table 28. Document revision history

Date	Version	Changes
08-Mar-2018	1	Initial release.
11-Oct-2018	2	Introduced STM32F417ZGT, hence added Section 5.3: STM32F417ZGT and the tables in it. Updated Table 5: <i>Safety results (by function) - Absolute metrics - STM32F469IIT</i> , Table 6: <i>Safety results (by function) - Relative metrics - STM32F469IIT</i> , Table 10: <i>Safety results (by function) - Absolute metrics - STM32F479BIT</i> and Table 11: <i>Safety results (by function) - Relative metrics - STM32F479BIT</i> .
10-Apr-2019	3	Introduced STM32F405VGT, hence added Section 5.4: STM32F405VGT and the tables in it.
25-Jun-2019	4	Updated functional safety documentation framework. Updated document title.
19-Sep-2023	5	Updated document title and Section Introduction. Added Section 4.4 Interpretation of FMEDA snapshot data. Added caution note in Section 1 General information. Updated Section 3.2 Reference documents and Section 4 FMEDA process and results. Corrected typo in Section 5.2 STM32F479BIT, Table 13. Safety results (MCU level) - Absolute metrics expressed in FITs - STM32F417ZGT and Table 18. Safety results (MCU level) - Absolute metrics expressed in FITs - STM32F405VGT. Added Section 5.5 STM32F410CBT.



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